

Ex.No.5: MIMO System Simulation with BER Analysis Date

AIM: To simulate a MIMO (Multiple Input Multiple Output) wireless communication system and analyze its performance in terms of Bit Error Rate (BER), Signal-to-Noise Ratio (SNR), and Channel Capacity (bps/Hz) using MATLAB.

Objective:

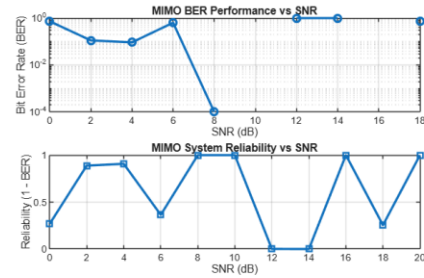
1. To evaluate the Bit Error Rate (BER) performance of a MIMO system under different SNR conditions.
2. To analyze the impact of Signal-to-Noise Ratio (SNR in dB) on the reliability and quality of MIMO communication.
3. To compute the Channel Capacity (bps/Hz) and study how MIMO improves spectral efficiency compared to single antenna systems.

Procedure

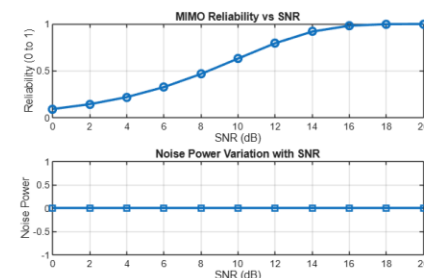
1. Define the MIMO system configuration (e.g., number of transmit and receive antennas such as 2×2 or 4×4).
2. Generate random binary data bits to represent the transmitted signal.
3. Modulate the bits using a digital modulation scheme (e.g., BPSK or QPSK).
4. Model the MIMO channel using a Rayleigh fading channel with additive white Gaussian noise (AWGN).
5. Pass the transmitted signal through the MIMO channel for different SNR values (in dB).
6. At the receiver, apply detection techniques (e.g., Zero-Forcing or Maximum Likelihood detection).
7. Demodulate the received signal to recover transmitted bits.
8. Compute Bit Error Rate (BER) by comparing transmitted and received bits.
9. Calculate Channel Capacity (bps/Hz) using SNR and MIMO capacity formula.
10. Plot BER vs SNR and Capacity vs SNR graphs for performance analysis.

Output

```
1 clc; clear; close all;
2 % SNR range
3 SNR_db = 0:2:20;
4 SNR = 10.^(SNR_db/10);
5 % MIMO parameters
6 Nt = 2; Nr = 2;
7 BER = zeros(1,length(SNR)); for i = 1:length(SNR)
8     bits = randi([0 1],10000,1);
9     symbols = 2*bits-1; % BPSK
10    H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
11    noise = (randn(Nr,10000) + 1j*randn(Nr,10000))/sqrt(2*SNR(i));
12    tx = H(1,1)*symbols + noise(1,:);
13    rx = real(tx) > 0;
14    BER(i) = sum(rx ~= bits)/length(bits);
15 end
16 figure
17 % ----- Graph 1: BER vs SNR -----
18 subplot(2,1,1)
19 semilogy(SNR_db,BER,'-o','linewidth',2)
20 title('MIMO BER Performance vs SNR')
21 xlabel('SNR (dB)')
22 ylabel('Bit Error Rate (BER)')
23 grid on
24 % ----- Graph 2: SNR vs Quality Indicator -----
25 subplot(2,1,2)
26 plot(SNR_db,1-BER,'-s','linewidth',2)
27 title('MIMO System Reliability vs SNR')
28 xlabel('SNR (dB)')
29 ylabel('Reliability (1 - BER)')
30 grid on
```

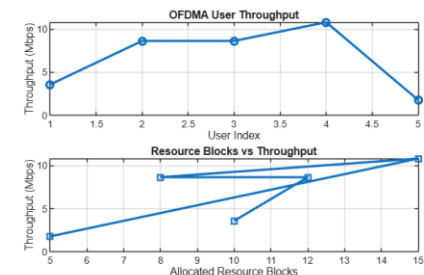


```
1 clc; clear; close all;
2 % SNR range (dB)
3 SNR_db = 0:2:20;
4 SNR = 10.^(SNR_db/10);
5 % MIMO parameters
6 Nt = 2; Nr = 2;
7 % Simulated reliability metric (inverse BER model)
8 Reliability = zeros(1,length(SNR));
9 for i = 1:length(SNR)
10     % simple channel model effect
11     H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
12     noise_power = 1/SNR(i);
13     % reliability increases with SNR
14     Reliability(i) = 1 - exp(-SNR(i)/10);
15 end
16 figure
17 % ----- Graph 1: SNR vs Reliability -----
18 subplot(2,1,1)
19 plot(SNR_db, Reliability, '-o', 'linewidth', 2)
20 title('MIMO Reliability vs SNR')
21 xlabel('SNR (dB)')
22 ylabel('Reliability (0 to 1)')
23 grid on
24 % ----- Graph 2: SNR vs Noise Effect -----
25 subplot(2,1,2)
26 plot(SNR_db, noise_power*ones(size(SNR_db)), '-s', 'linewidth', 2)
27 title('Noise Power Variation with SNR')
28 xlabel('SNR (dB)')
29 ylabel('Noise Power')
30 grid on
```



```
1 clc; clear; close all;
2 % Resource Block parameters
3 rb_bw = 180e3; % 180 kHz per RB
4 % Users
5 users = 1:5;
6 % Allocated Resource Blocks per user
7 alloc_RB = [10 12 8 15 5];
8 % Modulation efficiency (bits/symbol)
9 mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
10 % Throughput calculation (Mbps)
11 throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6; % Display results
12 disp('User Throughput (Mbps):')
13 disp(throughput)
14 figure
15 % ----- Graph 1: Throughput per User -----
16 subplot(2,1,1)
17 plot(users, throughput, '-o', 'linewidth', 2)
18 title('OFDMA User Throughput')
19 xlabel('User Index')
20 ylabel('Throughput (Mbps)')
21 grid on
22 % ----- Graph 2: RB vs Throughput -----
23 subplot(2,1,2)
24 plot(alloc_RB, throughput, '-s', 'linewidth', 2)
25 title('Resource Blocks vs Throughput')
26 xlabel('Allocated Resource Blocks')
27 ylabel('Throughput (Mbps)')
28 grid on
29
```

User Throughput (Mbps):
3.6000
8.6400
8.6400
18.8000
1.8000



Result

1. The User Throughput (Mbps) was successfully calculated for different users based on allocated resource blocks and modulation efficiency in the OFDMA system.
2. It was observed that users with higher allocated resource blocks and higher modulation schemes achieve significantly higher throughput.
3. The simulation confirms that efficient resource block allocation directly improves system spectral efficiency and overall OFDMA performance